

Radian measure



1

- a State the multiplier to convert an angle from degrees to radians.
- b State the multiplier to convert an angle from radians to degrees.

2 Convert the following to radians, correct to two decimal places where applicable:

- | | | | |
|-------------------|---------------|--------------|------------------|
| a 360° | b 180° | c 90° | d 225° |
| e -300° | f 112° | g 29° | h 161.17° |
| i $321^\circ 31'$ | j -45° | k 45° | l 330° |
| m -60° | n 120° | | |

3 Determine whether the following statements are true or false. If it is false, correct the statement.

- a If $180^\circ = \pi$ radians, then 90° must be equal to $\frac{\pi}{2}$ radians.
- b 60° must be equal to $\frac{\pi}{6}$ radians, because it is $\frac{1}{6}$ of 360° .
- c 30° must be equal to $\frac{\pi}{3}$ radians, because it is half of 60° .
- d 210° must be equal to $\frac{7\pi}{6}$ radians, because it is seven groups of 30° .
- e 45° must be equal to $\frac{\pi}{4}$ radians, because it is half of 90° .
- f 225° must be equal to $\frac{7\pi}{4}$ radians, because it is seven groups of 45° .

4 Convert the following to degrees:

- | | | | |
|--------------------|---------------------|---------------------|--------------------|
| a $\frac{\pi}{3}$ | b $\frac{\pi}{6}$ | c $\frac{\pi}{4}$ | d $\frac{3\pi}{2}$ |
| e $\frac{5\pi}{4}$ | f $-\frac{2\pi}{3}$ | g $-\frac{5\pi}{3}$ | h 4.2 radians |

5 For each of the following angles:

- | | | | |
|------------------------|---------------------------|------------------|----------------------------|
| i Find the complement. | ii Find the supplement. | | |
| a 30° | b $\frac{\pi}{4}$ radians | c $89^\circ 30'$ | d $\frac{\pi}{11}$ radians |



Unit circle

6 Determine the quadrant of the unit circle in which the following angles lie:

a 2.6 radians

b $-\frac{5\pi}{6}$ radians

c $\frac{\pi}{6}$ radians

d -6.4 radians

7 The graph shows a number of points on the unit circle:

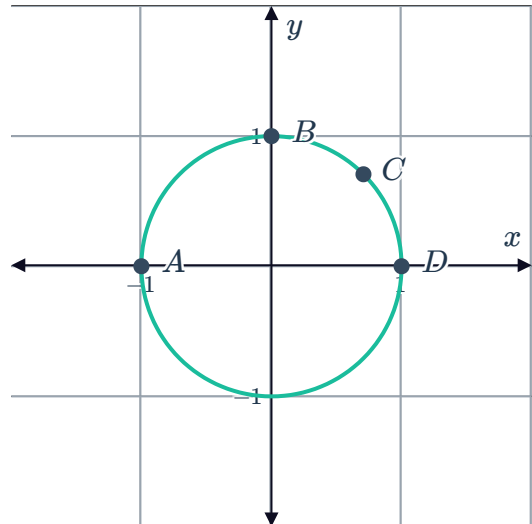
Name the point that corresponds to the following angle around the circle:

a $\frac{\pi}{2}$ units

b π units

c -2π units

d $-\frac{7\pi}{4}$ units



8 The graph shows a number of points on the unit circle:

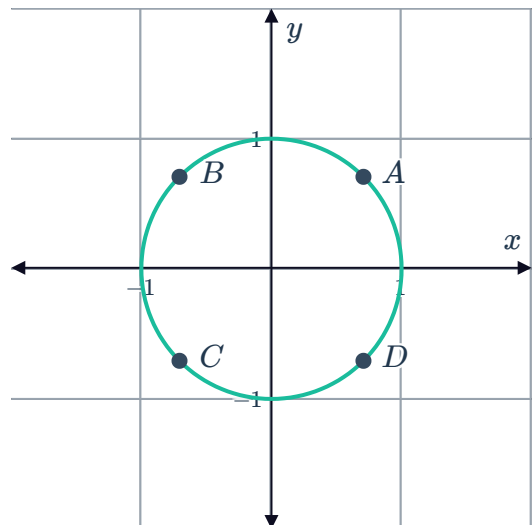
Name the point that corresponds to the following angle around the circle:

a $\frac{\pi}{4}$ units

b $\frac{3\pi}{4}$ units

c $\frac{21\pi}{4}$ units

d $-\frac{9\pi}{4}$ units



9 Given that π^c represents half a circle, state the fraction of the the unit circle represented by the following angles:

a $\frac{\pi^c}{2}$

b $\frac{2\pi^c}{3}$

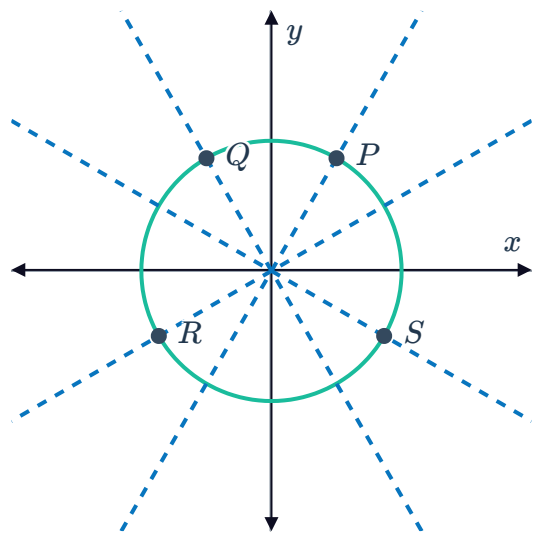
c $\frac{4\pi^c}{7}$

d $5\pi^c$

- 10 The graphs shows the unit circle divided into 12 equal segments, and a number of points on the circumference:
State two angles between -2π and 2π that correspond to the following points:



a R b P c S d Q



- 11 The graph shows a line through two points on the unit circle, A and B , and the origin, O . The line segment OB forms an angle of $\theta = \frac{\pi}{13}$ with the positive x -axis. The angle between the line segment OA and the positive x -axis is α .
Find the value of α when:

- a $0 \leq \alpha < 2\pi$
b $-2\pi < \alpha \leq 0$

